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1 Introduction

Climate change and it's representing one of the most pressing challenges faced by human societies today (IPCC, 2019). Average temperatures, as well as the frequency of intense weather events are projected to increase in the following decades, and at an accelerating rate (IPCC, 2019).

Faced with these trends, many municipalities around the world have adoped green roof by-laws (GRBLs) to mitigate against those climate-driven effects that are particularly relevant to urban centers (Berardi et al., 2014). These include the urban heat island effect, where metropolitan areas are warmer than their surrounding areas, in part due to lack of vegetated surfaces (Yang et al., 2016), and stormwater management (Jung et al., 2025). In addition to climate change preparedness, GRBLs often provide social services, by, for example, increasing resident access to green spaces (Berardi et al., 2014).

Toronto is the first city in North America to have had a By-law requiring the construction of green roofs (City of Toronto, 2026). From January 2010 to November 2025, all new developments and extensions with a floor area in excess of 2000 m² were required to construct a green roof, defined as “an extension of [a roof], built on top of a human-made structure, that allows vegetation to grow in a growing medium” (City of Toronto, 2026). Complimenting their GRBL, the City also implemented an Eco-Roof Incentive Program (ERIP) to support the voluntary installation of green roofs on homes and buildings (City of Toronto, 2026).

In this report, City building permit registry data was leveraged to document and analyze the state of Toronto's green roofs. In particular, we were interested in understanding (i) where and on what types of structures green roofs are (planned to be) installed, and (ii) the past and current state of green roof adoption in Toronto. I find that [summary of results]. The remainder of the report is structured as follows. Section 2 [hyperlink] ...

*Code and data are available at: https://github.com/bennypk1/starter_folder.

2 Data

2.1 Overview

The dataset used in this report represents an up-to-date collection of 1235 building permits related to Toronto’s GRBL (n=1194) and ERIP (n=41). Both the GRBL and ERIP were adopted in 2009 by the Toronto City Council, and entered into effect on January 31, 2010. The GRBL was officially repealed in November 2025, making green roofs voluntary for private developers (City of Toronto, 2026). Permit information (described in more detail below) of all structures for which the GRBL or ERIP has applied are made public and updated weekly on the City’s Open Data website (City of Toronto, 2026), accessed for this report on May 6, 2026. The raw dataset also contained an additional 15 definitively cancelled projects, which were removed [in one sentence, why].

Each permit is assigned a Permit Number, which consists of the last two digits of its application year (e.g. 11 for 2011), followed by a 6-digit IBMS-generated sequence. After issuance, permit applications may be revised, the proposed changes of which may qualify for GRBL or ERIP. Such revisions are recorded using the same Permit Number, and distinguished from the original permit application by an integer value (called the Recording Number). Therefore, identifying truly unique green roofs requires specifying both a Permit and Revision Number (this one-to-one mapping was violated once in the dataset ; the structure’s duplicate observations were removed and assumed to be errors). Permit-revision pairs are each assigned types (e.g. “New Building”, “Small Residential Projects”), a current status (e.g. “Closed”, “Revision Issued”), and a written description of the proposed project. Dates of permit application, issuance, and completion are also provided where applicable. In addition, the full address, forward sortation area (FSA ; i.e. the first three digits of the postal code), type (e.g. “University”, “SFD – Detached”), and green roof size (in m2) of the associated structure is provided. All data acquisition, cleaning, tabulation, and visualization tasks were conducted within the R Statistical Software (R Core Team, 2023). Notable packages used in this report include `opendatatoronto` (Gelfand, 2022), `tidyverse` (Wickham et al., 2019), and `canadianmaps` (Cayen, 2024).

2.2 Outcome variables

Add graphs, tables and text. Use sub-sub-headings for each outcome variable or update the subheading to be singular.

Talk more about it.

And also planes (**?@fig-planes**). (You can change the height and width, but don’t worry about doing that until you have finished every other aspect of the paper - Quarto will try to make it look nice and the defaults usually work well once you have enough text.)

Talk way more about it.

2.3 Predictor variables

Add graphs, tables and text.

Use sub-sub-headings for each outcome variable and feel free to combine a few into one if they go together naturally.

3 Results

Our results are summarized in ?@tbl-modelresults.

4 Discussion

4.1 First discussion point

If my paper were 10 pages, then should be at least 2.5 pages. The discussion is a chance to show off what you know and what you learnt from all this.

4.2 Second discussion point

Please don't use these as sub-heading labels - change them to be what your point actually is.

4.3 Third discussion point

4.4 Weaknesses and next steps

Weaknesses and next steps should also be included.

Appendix

A Additional data details

B Model details

B.1 Posterior predictive check

In `?@fig-ppcheckandposteriorvsprior-1` we implement a posterior predictive check. This shows...

In `?@fig-ppcheckandposteriorvsprior-2` we compare the posterior with the prior. This shows...

B.2 Diagnostics

`?@fig-stanareyouokay-1` is a trace plot. It shows... This suggests...

`?@fig-stanareyouokay-2` is a Rhat plot. It shows... This suggests...

C References